

Gas Absorption in a Packed Bed

Group #2

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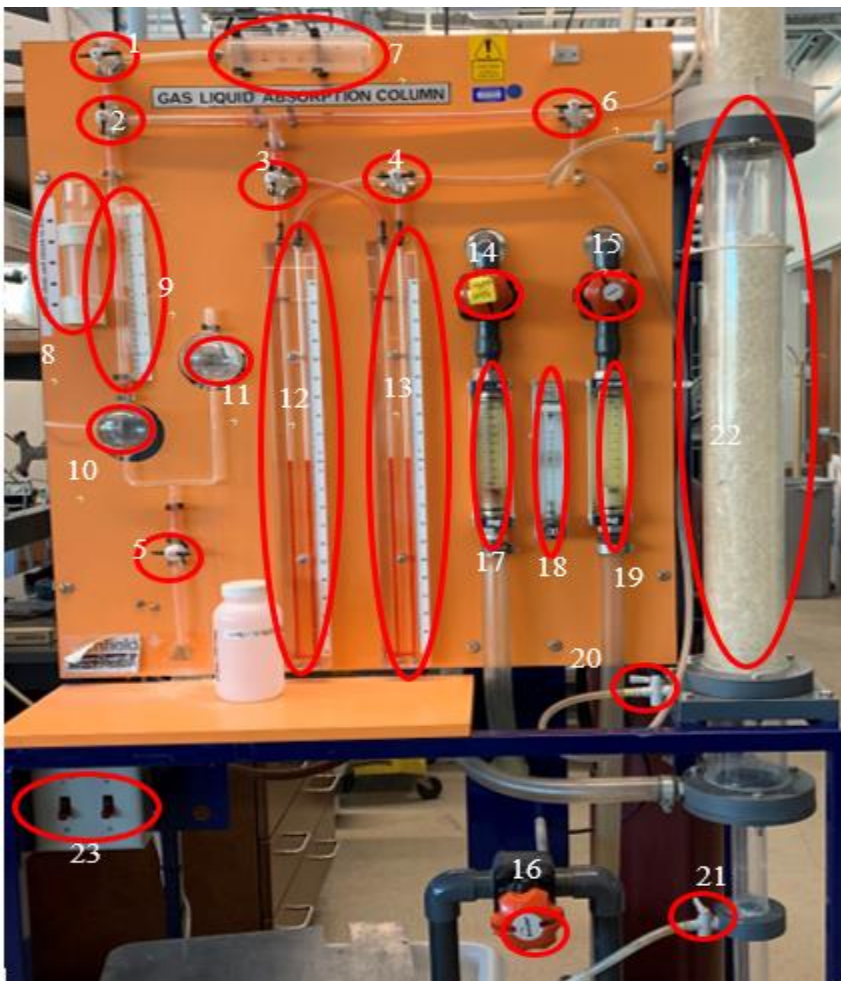
Introduction

- The purpose of this experiment is to determine fluid flow and gas absorption components in the removal of CO₂ from air stream with water in a packed bed.
- **Components**: Flooding Point, overall mass transfer coefficient, CO₂ Concentration
- **Parameters**: Water flowrate, air flowrate, CO₂ flowrate, control valves
- **Key**: Titration – Go slow, any overshooting will create fluctuations in data. Measure sample's volume for more accurate results. Cover samples with parafilm to avoid escaping CO₂

Theory

- The tower, which is packed with Raschig beads, has a counter-current flow. The water is pumped into the top of the column, where the gas is pushed into the bottom.
- Flooding occurs when the column becomes overrun with water. This will cause the water to start to flow into the gas pipes, which can cause serious damage to the equipment.
- Void Fraction
 - $\varepsilon = \frac{V_{void}}{V_{apparent}}$ or $\varepsilon = \frac{\rho_B}{\rho_S}$
- CO₂ Concentration
 - $C_{CO_2} = \frac{V_{B^*}(C_B)}{V_{samp}}$
- Mass Transfer Coefficient
 - $N = K * a * \Delta C$
- McCabe Thiele
 - *Steady State* = d(GMy) = d(LMx)
 - *OC Differential* = $GM' \frac{dy}{(1-y)^2} = LM' \frac{dx}{(1-x)^2}$
 - $GM' \left[\left(\frac{y}{1-y} \right) - \left(\frac{y_2}{1-y_2} \right) \right] = LM' \left[\left(\frac{x}{1-x} \right) - \left(\frac{x_2}{1-x_2} \right) \right]$
- Henrys Law
 - $p_i = He[A]$

Apparatus



No.	Name
1-6	Three-Way Valves
7	Volume Measuring Tube
8	Plunger
9	NaOH Measurement
10-11	Bulbs
12-13	Manometer
14	“Do Not Touch” Valve
15-16	Variable Valve
17	Air Flow Meter
18	Carbon Dioxide Flow Meter
19	Water Flow Meter
20-21	Go/No-go Valves

Apparatus



No.	Name
22	Gas Absorption Column
23	Switches
24	Go/No-Go Valves
25	Heater
26	Regulator Valve
27	CO2 Tank
28	Pump
29	Water Tank
30	Regulator Valve
31	Go/No-Go

Material & Supplies

- 50mL burette with clamp
- 50mL graduated cylinder
- 125mL Erlenmeyer flasks
- Thermometer
- CO₂
- Compressed air
- DI water
- NaOH - 0.5M
- Parafilm
- Phenolphthalein

Procedure

- **Part 1: Flooding** – Fill tank with DI water. Adjust air flow rate to 20L/min to dry column, then start pump and adjust water flow to 1.5L/min. Measure pressure drops with increasing air flow until flooding occurs. Repeat with 3, 6, & 8L/min water. Measure pressure drops at each stage
- **Part 2a: Absorption** – Set air flow to 20L/min, water flow at 6L/min, and CO₂ flow to what's assigned. Take 50mL samples and analyze CO₂ concentration until constant with titrations (1min intervals)
- **Part 2b: Desorption** – From 2a, set water flow to 1.5L/min, open drain valve, and shut off CO₂, and measure CO₂ concentration with titrations (1min intervals)
- **Part 3: Continuous** – Drain water tank and refill again. Same as 2a but leaving drain valve open and water flow rate at 1.5L/min. Take samples every 30 seconds, 1 min, and 5 min intervals

Experimental Challenges

- Collecting Samples - Make sure you measure volume once done taking sample, helps with titration data
- Need to move fast! - Portion of the lab where drain must be open, tank empties quickly. Can get hose in flow more water in tank

Safety

- Electrical Hazard
 - The pumps when water is in contact may cause a fire or mechanical failures
- Pressure Hazard
 - High pressure gas exceeding allowed temperature may cause mechanical failure.
- Temperature Hazard
 - Water can increase temperature throughout the lifespan of the system running, therefore replacing the water after experiment 1 is required.
- Chemical Hazard
 - NaOH is flammable and may cause eye damage and skin burns.
- Gloves are required while handling the NaOH and while titrating
- Goggles to be worn at all times
- Leave control valve for air flow rate open